Session Objectives

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At the conclusion of the session, the student should be able:

1. Define & describe the factors influencing K^+ balance.
2. Define & describe the renin-angiotensin-aldosterone system and its role in K^+ balance in the nephron.
3. Interpret the relationships between acid/base balance and K^+ homeostasis.
4. Describe the regulation of K^+, Ca^{++}, Mg^{++} & PO_4^{3-} in the nephron.

The Nephron

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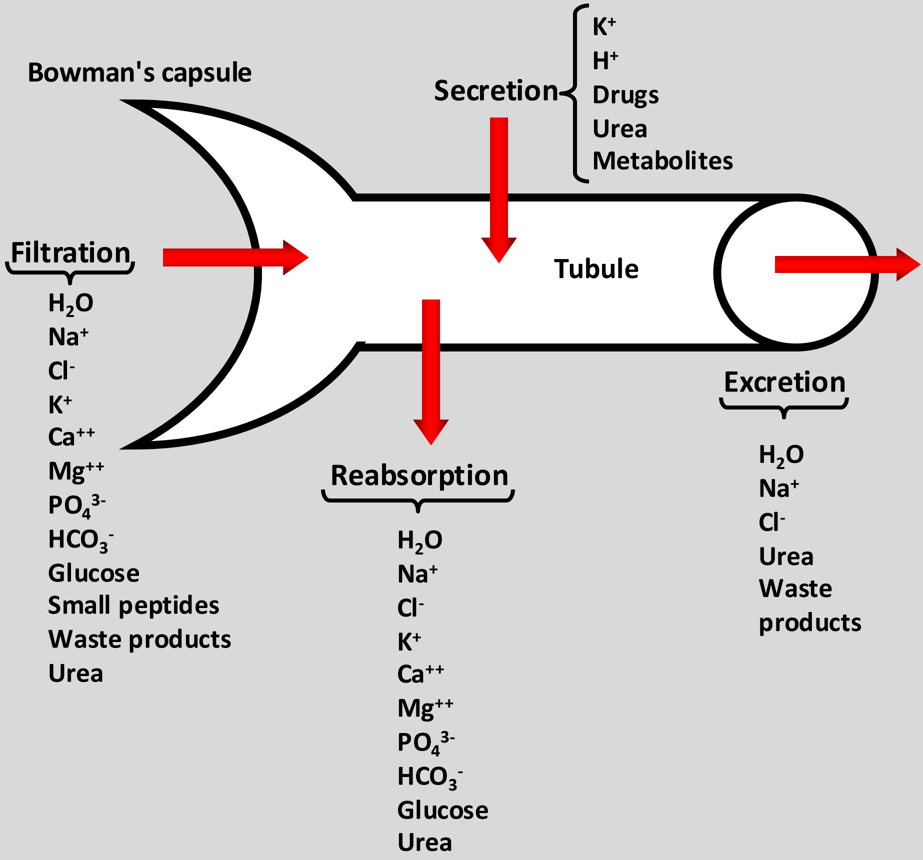
- The kidney plays a central role in controlling the plasma levels of a wide range of solutes
- All are present at low concentrations in the body
- The renal excretion of a solute depends on three processes: Filtration, Reabsorption & Secretion
- The kidney filters and then reabsorbs substances, some totally (e.g., glucose).
- It also filter & secrete (e.g., K^+, organic anions & cations)
- It can also filter, reabsorb, and secrete (e.g., urea).
- The rates at which different substances are excreted represent the sum of these three renal processes

Urinary excretion rate = Filtration rate − Reabsorption rate + Secretion rate

- In general, tubular reabsorption is quantitatively more important than tubular secretion in urine formation
- Secretion plays an important role in determining the amounts of K^+, H^+ & other substances.
- Electrolytes (e.g. Na^+, Cl^-, HCO_3^-) are highly reabsorbed.

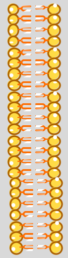
NEPHRON



K^+ Balance & Renal Regulation of K^+

- The distribution of K^+ differs from that of Na^+
- ~98% of the total-body K^+ is ICF vs 2% in the extracellular fluid (ECF).
- Plasma $[K^+]$ is tightly maintained at 3.5-5.0 mM.
- Chronic hypokalemia leads to several metabolic disturbances:
  - Inability of the kidney to form concentrated urine
  - Metabolic alkalosis
  - Enhancement of renal excretion of NH_3
- Elevated $[K^+]_{IC}$ is important for:
  - Maintenance of cell volume
  - Enzyme function
  - DNA and protein synthesis
  - Cell growth
- Low $[K^+]_{EC}$:
  - <u>Integral for membrane potential of excitable and non-excitable cells.</u>
- Changes in $[K^+]$ ratios can cause disturbances in excitation and contraction of muscles.
- Doubling the plasma $[K^+]_{Pl}$/Reduction by half:
  - Severe disturbances in skeletal and cardiac muscle function.
  - Potentially life-threatening disturbances of cardiac rhythmicity from hyperkalemia

Plasma Membrane



ECF/Plasma	Intracellular Fluid
$[Na^+] \cong 142$ mM	$[Na^+] \cong 15$ mM
$[K^+] \cong 4.4$ mM	$[K^+] \cong 125$ mM
$[Ca^{++}] \cong 2.4$ mM	$[Cl^-] \cong 20$ mM
$[Mg^{++}] \cong 1$mM	$[Mg^{++}] \cong 20$ nM
$[PO_3^-] \cong 1$mM	$[PO_3^-] \cong 5$-70 mM
$[Cl^-] \cong 102$ mM	$[Ca^{++}] \cong 100$ nM

Intracellular $[K^+]$ ($[K^+]_{IC}$)		
	$\downarrow [K^+]_{IC}$	$\uparrow [K^+]_{IC}$
Cell volume	Shrinkage	Swelling
DNA/protein synthesis, growth	Reduction	+/-

Extracellular$[K^+]$ ($[K^+]_{EC}$)		
	$\downarrow [K^+]_{EC}$	$\uparrow [K^+]_{EC}$
pH_{IC} regulation	Acidosis	Alkalosis
Vascular tone	Constriction	Relaxation
Neuromuscular Activity	Muscle weakness & paralysis, resp. failure, abdominal distension	Increase excitability, eventual flaccid paralysis
Cardiac activity	prolonged repolarization, slowed conduction, abnormal pacemaker activity, tachyarrhythmia	enhanced repolarization, slowed conduction, bradyarrhythmia,

Intracellular $[K^+]$/Extracelluar $[K^+]$ ($[K^+]_{IC}/[K^+]_{EC}$)		
	$\downarrow$($[K^+]_{IC}/[K^+]_{EC}$)	$\uparrow$($[K^+]_{IC}/[K^+]_{EC}$)
Resting membrane Pot.	Depolarization	Hyperpolarization

K^+ Homeostasis

- Homeostasis involves external K^+ (environment vs. body) & internal K^+ balance (IC vs EC)
- External K^+ Balance:
  - Determined by the relationship between $[K^+]$ intake & excretion.
  - $[K^+]$ intake = $[K^+]_{EC}$ (~ 65-75 mmol)
  - For $[K^+]_{EC}$ to remain constant, the body must excrete $[K^+]$ via renal & extrarenal mechanisms at the same rate as ingestion.
  - $[K^+]$ excretory mechanisms must be able to adjust to variable intake.
  - The kidney is largely responsible for $[K^+]$ excretion
  - GI tract plays a minor role.
  - GI tract can adjust $[K^+]$ excretion in response to:
    - Adrenal hormones
    - Δ in $[K^+]$ intake
    - Decreased capacity of the kidneys to excrete $[K^+]$.
- Internal K^+ Balance:
  - Majority of total body $[K^+]$ is intracellular.
  - Unequal distribution between IC & EC $[K^+]$ has important implications.
  - Total $[K^+]$ ~ 3000 mmol
  - Shuttling as little as 1% to or from the ECF would cause a 50% change in extracellular $[K^+]$
  - Severe consequences for neuromuscular function

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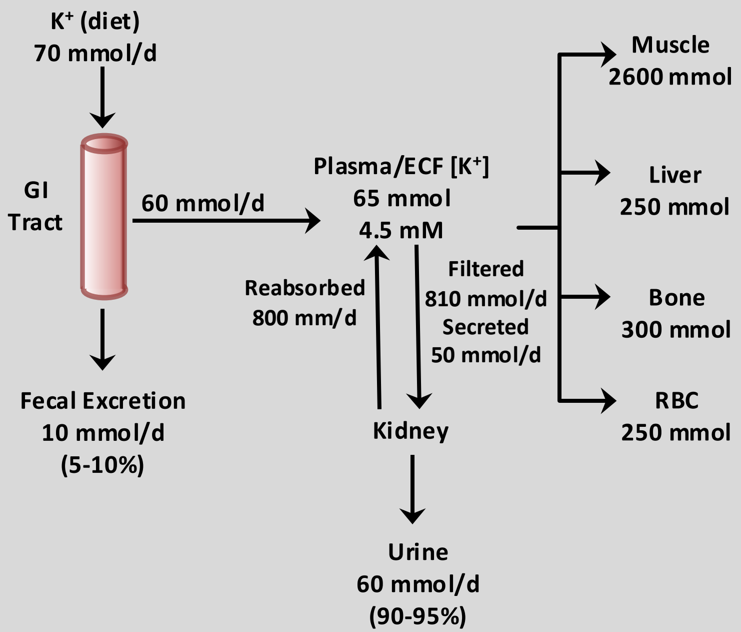
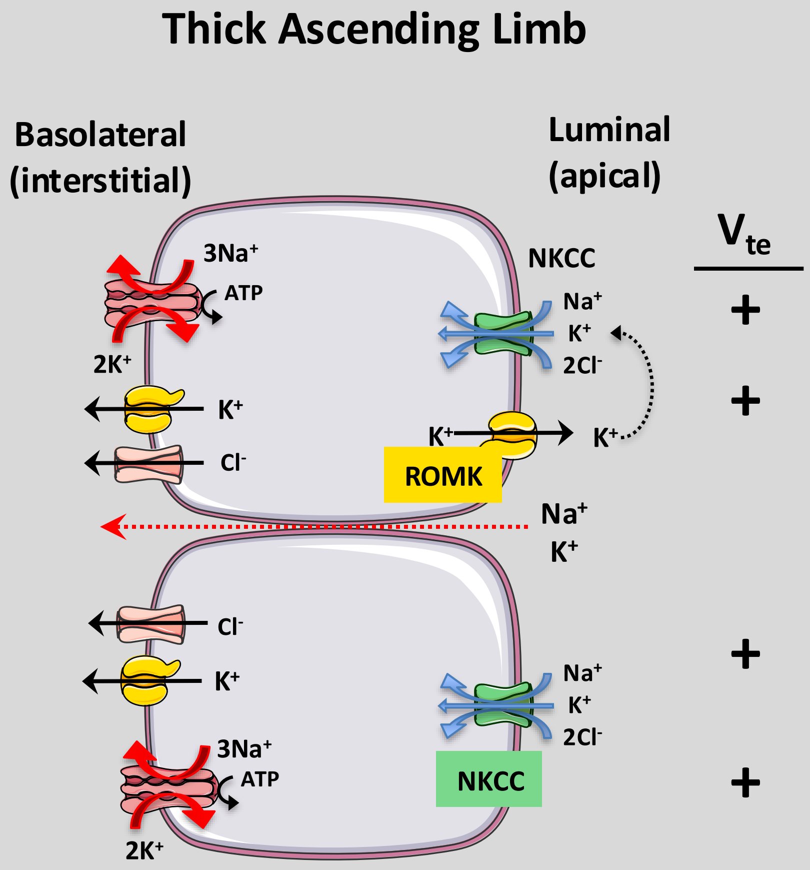
K^+ Distribution & Balance



K^+ Reabsorption in the Thick Ascending Limb

- The <u>descending</u> limb of the loops of Henle (tDLH) *secrete* K^+.
- Secretion is passive, driven by the high [K^+] of the medullary interstitium & elevated [K^+] permeability.
- K^+ Reabsorption in the tALH:
  - Paracellular pathway:
    - Passive reabsorption
    - Driving force is the lumen-to-interstitium concentration gradient.
    - Minor pathway.
  - Transcellular pathway:
    - Majority of K^+ use the apical $Na^+/K^+/2Cl^-$ co-transporter.
    - Removing any of the ions from the lumen abolishes reabsorption.
    - Inhibition of NKCC abolishes the lumen-positive V_{te}.
    - This will also block paracellular K^+ reabsorption.
- NKCC is sensitive to high-ceiling class of diuretics (Loop Diuretics).
- Apical K^+ conductance occurs via ROMK channels.
- Major function of ROMK channel:
  - Mechanism for K^+ recycling



Nephron Segment	Transepithelial Voltage (mV)
PCT (S1)	-3
PCT (S3)	+3
TAL	+6
DCT	-24 to -14
CCT	-74

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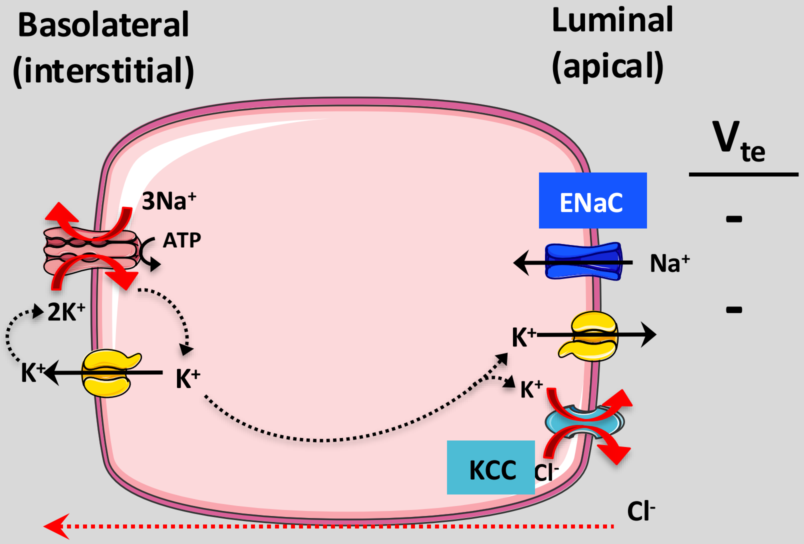
K^+ Reabsorption in the Collecting Tubule

- Urinary K^+ content depends on the distal K^+ secretory system (DCT, CT, CCT & MCD).
- Ultrastructurally, the ICT and CCT are nearly identical
- They are made up of:
  - Principal cells (~70%):
    - Secrete K^+ into the tubular fluid
    - Epithelial Na^+ channel (ENaC)
  - Intercalated cells (~30%): Some K^+ reabsorption from the tubular fluid
- Early portion of the DCT secrete K^+ at a low rate.
- Late portions of the classic distal tubule (i.e., CT, CCT) have a high rate of K^+ secretion.
- Principal Cell:
  - Mainly responsible for K^+ secretion
  - The three key elements:
    - Na^+-K^+ ATPase for K^+ (basolateral membrane).
    - High & variable apical K^+ permeability (ROMK channel)
    - EC driving force for K^+ exit across the apical membrane.
    - K^+ can also move to lumen via an apical K^+/Cl^- cotransporter (KCC).
- Net effect: Active movement of basolateral K^+ $\rightarrow$ cell $\rightarrow$ passive movement to lumen (urine).
- Changes in any of the three elements can affect the secretion of K^+.

Collecting Tubule
Principal Cells



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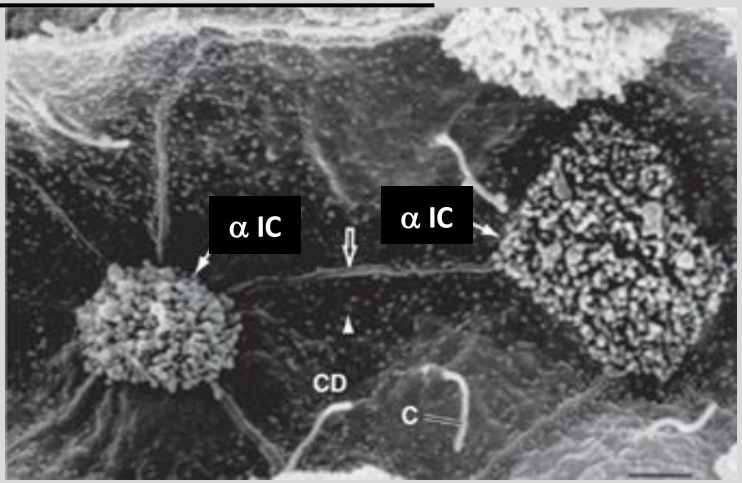
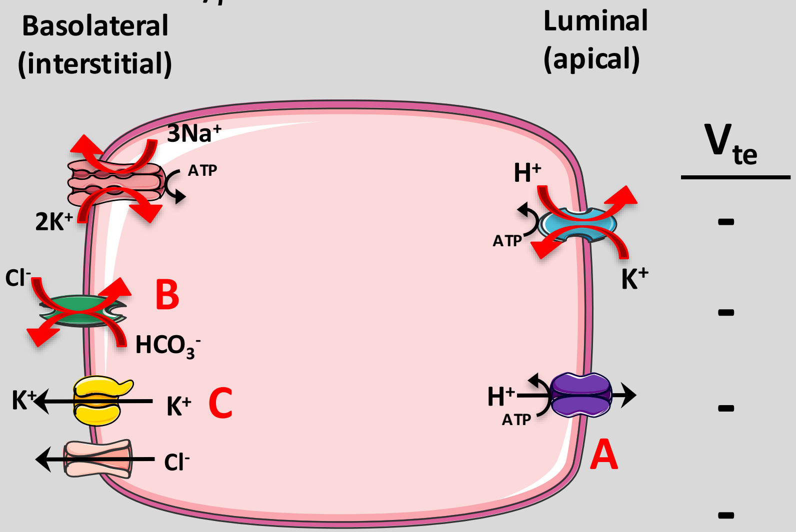
K^+ Reabsorption in the Collecting Tubule

- Intercalated cells:
  - Type A (α) Intercalated Cells: 1° role in H^+ secretion (A)
  - Type B (β) Intercalated Cells: 1° role in HCO_3^- reabsorption (B)
  - Morphologically distinct from the Principal cells
  - ICT, CCT, and MCD reabsorb K^+ in response to K^+ depletion (transcellular reabsorption).
- Reabsorption occurs in 2 steps:
  - Active step mediated by an apical H^+/K^+ ATPase (A)
  - Passive step mediated by a basolateral K^+ channel (allows leakage) (C)
- K^+ depletion produces an increase in H^+/K^+ ATPase density & activity.
- This will enhance K^+ reabsorption.
- Increase in H^+/K^+ ATPase activity accelerates H^+ secretion
- This contributes to the development of <u>HYPOKALEMIC ALKALOSIS</u>.



Collecting Tubule

α/β-Intercalated Cells



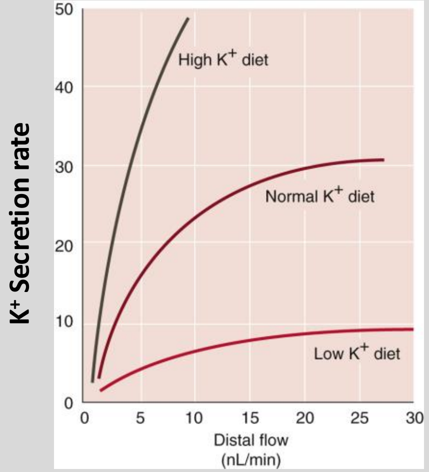
Nephron Segment	Transepithelial Voltage (mV)
PCT (S1)	-3
PCT (S3)	+3
TAL	+6
DCT	-24 to -14
CCT	-74

Factors Modulating Secretion by the Distal K^+-Secretory System

- K^+ reabsorption along the MCD is both passive and active.
- Capacity for K^+ secretion diminishes from cortex to MCD.
- K^+ loss from the MCD lumen can occur by passive movement via the paracellular pathway driven by $[K^+]$ gradient.
- Luminal $[K^+]$ of the MCD is high for:
  - Upstream segments (distal K^+-secretory system) secrete K^+.
  - Fluid reabsorption of fluid in the medullary collecting tubules in the presence of high arginine vasopressin (AVP) levels increase luminal K^+.
- In addition to passive K^+ reabsorption, apical H^+/K^+ ATPase mediate K^+ reabsorption, especially during low K^+ intake.
- Fluid flow:
  - Very potent stimulus of K^+ secretion along the ICT and CCT.
  - Similar relationship holds between urine flow and K^+ excretion (kaliuresis).
  - Increased urine flow (e.g., EC volume expansion, osmotic diuresis, or administration of diuretic agents) $\rightarrow$ Enhanced kaliuresis.
- Increased luminal flow also increases Na^+ in tubule cells
- This $\uparrow$luminal $[Na^+]$ and enhances Na^+ uptake by principal cells
- This stimulates H^+/K^+ ATPase activity $\rightarrow$ $\uparrow K^+$ secretion.
- $\uparrow$urinary flow is associated with increased natriuresis

LUMINAL FACTORS	
STIMULATORS	INHIBITORS
$\uparrow$ Flow Rate	$\uparrow [K^+]$
$\uparrow [Na^+]$	$\uparrow [Cl^-]$
$\downarrow [Cl^-]$	$\uparrow [Ca^{++}]$
Negative luminal voltage	Barium
Diuretics acting upstream	Amiloride

PERITUBULAR FACTORS	
STIMULATORS	INHIBITORS
$\uparrow K^+$ Intake	$\downarrow$pH
$\uparrow [K^+]$	Epinephrine
$\uparrow$ pH	
Mineralocorticoids	
Arginine vasopressin	



Transepithelial Potential & K^+ Secretion

- K^+ secretion in the ICT and CCT occurs by diffusion of K^+ from the principal cell to the lumen.
- Process depends on the apical electrochemical K^+ gradient.
- Increases in luminal [Na^+] enhance apical Na^+ entry through epithelial Na^+ channels (ENaCs).
- This ***depolarizes*** the apical membrane, favoring the exit of K^+ from cell to lumen.
- Conversely, a fall in luminal [Na^+] will ***hyperpolarize*** the apical membrane
- This will inhibit K^+ secretion.
- K^+-sparing diuretics (e.g., amiloride) block apical ENaCs
- Same effect on K^+ secretion as decreasing luminal [Na^+].
- This hyperpolarizes the apical membrane & reduces E.C. gradient for K^+ secretion.
- Inhibition of apical Na^+ will lower ↓[Na^+] and reduce Na^+-K^+ ATPase activity.
- Lowering luminal [Cl^-] & replacing it with poorly-reabsorbed anion (e.g., SO_4^{-2}, HCO_3^-) promotes urinary K^+
- This is independent of changes in the lumen-negative V_{te}.
- Lowering luminal [Cl^-] increases the cell → lumen Cl^- gradient
- This stimulates K^+/Cl^- cotransporter (KCC) and K^+ secretion.

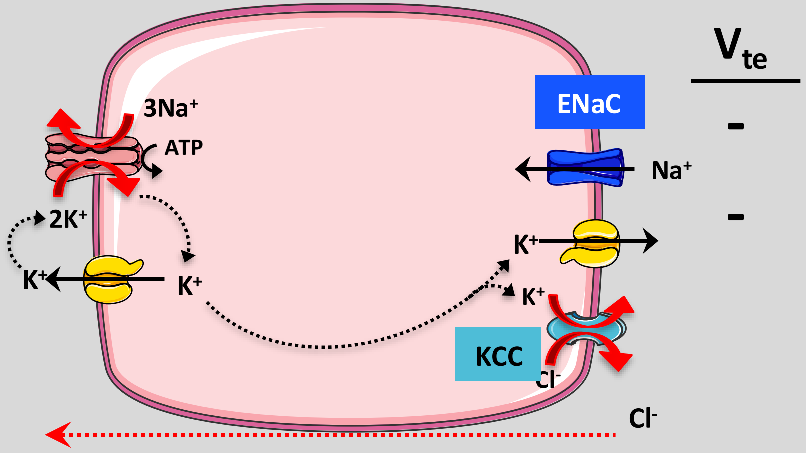
LUMINAL FACTORS	
STIMULATORS	INHIBITORS
↑ Flow Rate	↑ [K^+]
↑ [Na^+]	↑ [Cl^+]
↓ [Cl^-]	↑ [Ca^{++}]
Negative luminal voltage	Barium
Diuretics acting upstream	Amiloride

PERITUBULAR FACTORS	
STIMULATORS	INHIBITORS
↑ K^+ Intake	↓pH
↑ [K^+]	Epinephrine
↑ pH	
Mineralocorticoids	
Arginine vasopressin	

Principal Cells

Basolateral (interstitial)

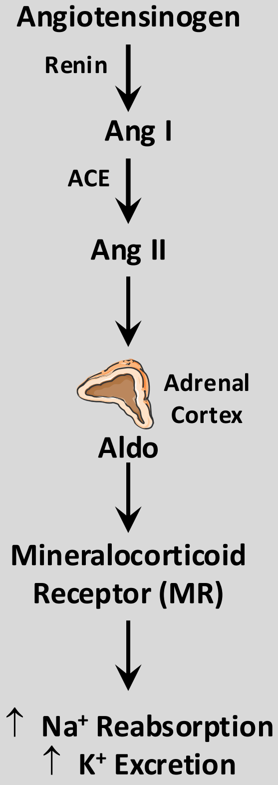
Luminal (apical)

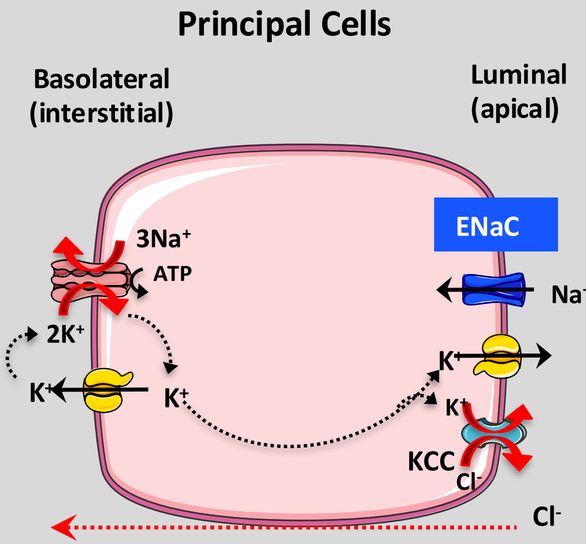


Corticosteroids & K^+ Secretion

- The adrenal cortex is the major site of steroid hormone synthesis (glucocorticoids (GC), mineralocorticoids (MC), adrenal androgens (17-ketosteroids), estrogens, and progestogens).
- Aldosterone (Aldo) is the MAJOR mineralocorticoid receptor (MR) produced.
- Secretion of Aldo from the zona glomerulosa of the adrenal cortex is regulated by Ang II, K^+, and (to a lesser extent) ACTH.
- The renin-angiotensin-aldosterone system (RAAS) is the major regulator of renal sodium excretion and blood pressure.
- Aldo induces K^+ secretion in the ICT and CCT via the transcription of ENaC and ROMK.
- Aldo also increases K^+ uptake by stimulating Na^+-K^+ ATPase activity & #.
- Extent to which Aldo increases K^+ excretion depends on Na^+ excretion (e.g. low Na^+ intake and excretion)
- Under physiological conditions, GC enhance K^+ excretion.
- Natural & synthetic GC (corticosterone & dexamethasone, respectively) produce this by increasing flow along the ICT and CCT.
- GCs also increase GFR, increasing amounts of H_2O & Na^+ in these segments.
- Unphysiologically high [GC] stimulate K^+ by binding & activating MR's.

PERITUBULAR FACTORS	
STIMULATORS	INHIBITORS
↑ K^+ Intake	↓pH
↑ $[K^+]$	Epinephrine
↑ pH	
Mineralocorticoids	
Arginine vasopressin	





Renal Regulation of Ca^{++}

- Ca^{++} plays a ***critical*** role in secretion, contraction, conduction, exocytosis, and the activation and inactivation of enzymatic pathways.
- $[Ca^{++}]_{Pl}$ is strictly regulated
- Protein-bound Ca^{++}: Bound to plasma proteins (mostly albumin) & are nonfilterable.
- The filterable portion: Diffusible Ca^{++} complexes (bound to CO_3^{-2}, citrate, PO_4^{3-} & SO_4^{3-}) & free, ionized Ca^{++}.
- Free, ionized Ca^{++} is tightly regulated.
- Distribution of various forms of total calcium is not fixed.
- H^+ competes with Ca^{++} for binding to anionic sites on proteins or small molecules
- Acidosis: Increases plasma $[Ca^{++}]_{Pl}$
- Alkalosis: Acutely reduces $[Ca^{++}]_{Pl}$
- Acute alkalosis can cause signs and symptoms mimicking hypocalcemia.
- Ca^{++} binding to plasma proteins depends on the concentrations of the proteins.
- 99% is stored in bone
- Chemically linked with other components of hydroxyapatite $[Ca_5(PO_4)_3(OH)]$
- Hypercalcemia: Multiple effects (effects on excitable cell function)
- Hypercalcemia: CNS depression, muscle weakness, and GI tract immobility.

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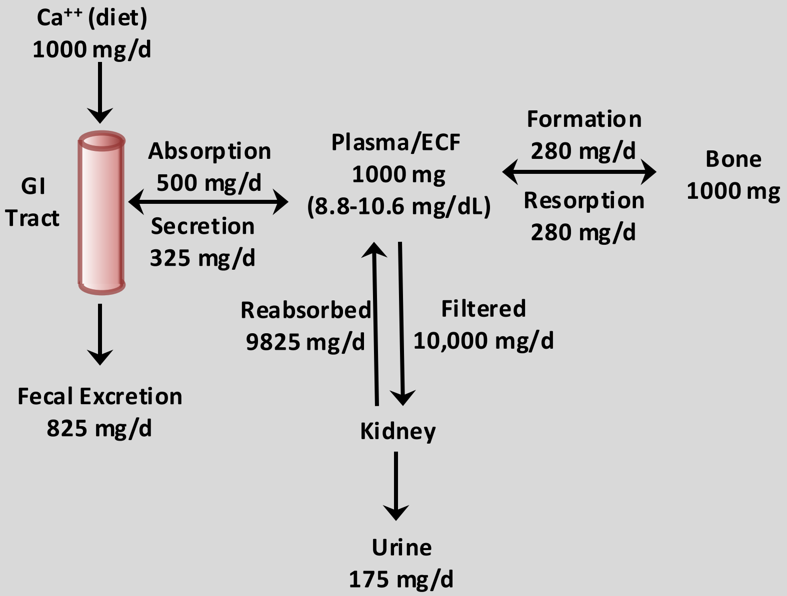
Ca^{++} Distribution & Balance



	mg/dL	mM	% Total
Ionized Ca^{++}	4.7	1.2	~ 45
Diffusible Ca^{++} complexes	1.4	0.3	~15
Protein-bound Ca^{++}	3.9	1.0	~40
Total Ca^{++}	10.0	2.5	100

Ca^{++} Transport in the Nephron

- 99% of the filtered Ca^{++} is reabsorbed.
- Occurs in the PCT, tALH & DCT
- PCT & TALH: Reabsorption is tightly coupled to Na^{+} reabsorption
- DCT: Reabsorption is dissociated from Na^{+}.

Proximal Tubule:

- Reabsorbs ~ 65% of the filtered calcium (NOT SUBJECT TO HORMONAL CONTROL).
- All PCT Ca^{++} reabsorption occurs via the paracellular routes.

PCT:

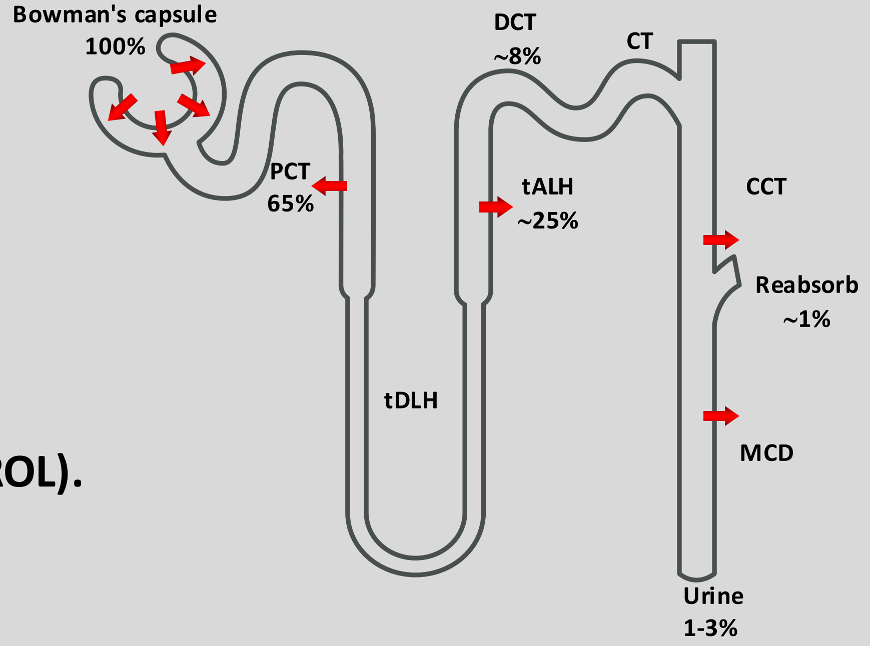
- High Ca^{++} permeability
- Transport is sensitive to changes in V_{te}
- Lumen-negative V_{te} → Ca^{++} secretion
- Lumen-positive V_{te} → Ca^{++} reabsorption.
- Paracellular reabsorption: Dependent on luminal $[Ca^{++}]$.

H_2O reabsorption → ↑ $[Ca^{++}]_L$ → favorable V_{te} → Ca^{++} reabsorption.

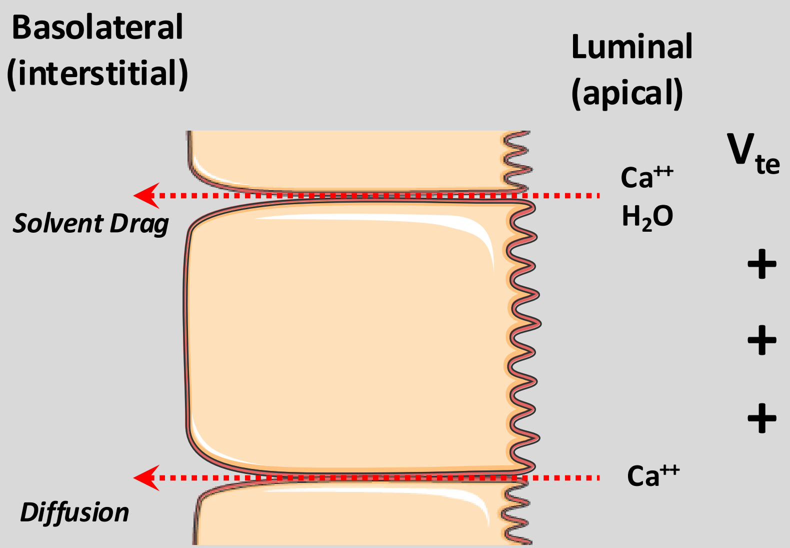
- The greater the fluid reabsorption along the PT, the greater the reabsorption

↑ NaCl diet → ↑ECV → ↓ Na^{+} & H_2O reabsorption → ↓ Ca^{++} reabsorption→ calciuria

Ca^{++} Homeostasis in the Nephron



Proximal Tubule



Ca^{++} Transport in the Nephron

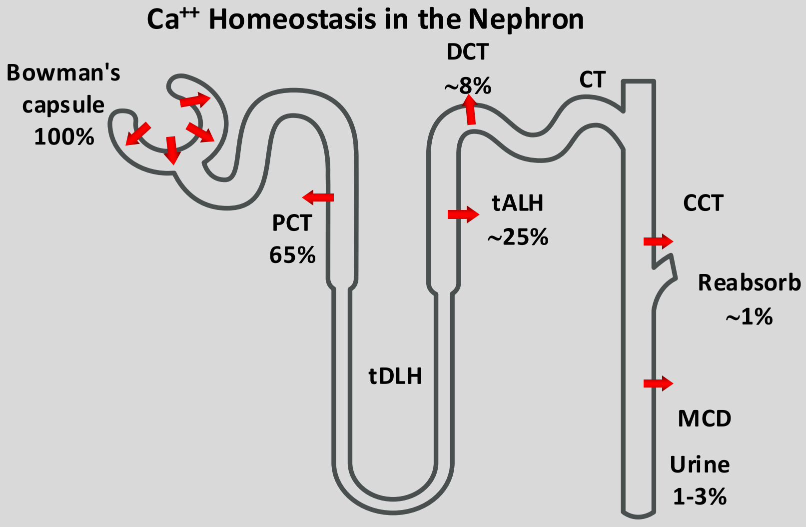
tALH:

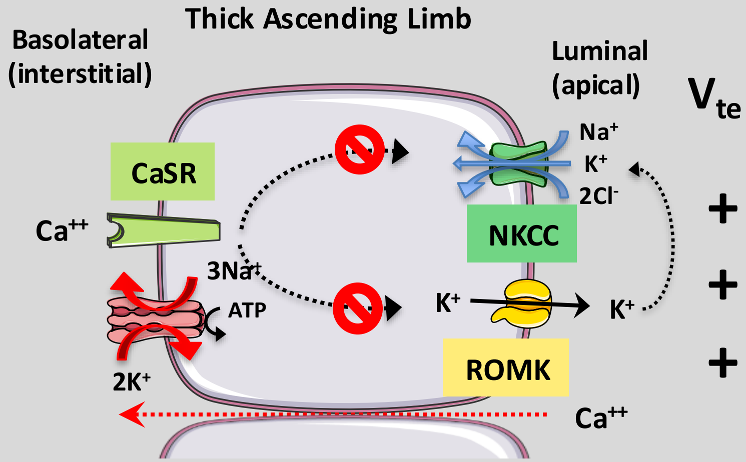
- Reabsorbs ~ 25% of the FL of Ca^{++}.
- Most of the Ca^{++} reabsorption occurs passively via a paracellular route
- Driven by the lumen-positive V_{te}.
- $[Ca^{++}]$ can also bind to a basolateral Ca^{++}-sensing receptor (CaSR)
- Results in inhibition of ROMK & NKCC → ↓lumen-positive V_{te}
- ↓Ca^{++} reabsorption & and ↑ Ca^{++} excretion.

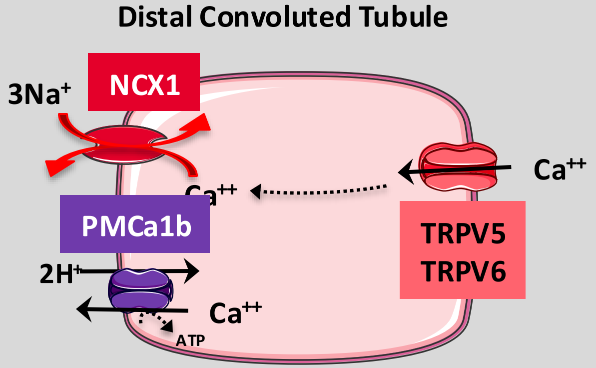
DCT:

Reabsorbs ~ 8% of the FL of Ca^{++}

- Major regulatory site despite small amount of Ca^{++}.
- Reabsorption is predominantly via active, transcellular route.
- MP of DCT ~ -70 mV (steep electrochemical gradient)
- Apical Ca^{++} is passive, mediated by the epithelial Ca^{++} channels (TRPV5 & TRPV6).
- DCT cells must extrude across the basolateral membrane all Ca^{++} entering cells
- Both 1° (Ca^{++}-ATPase, PMCa1b) & 2° active transporters participate.
- The Na^+-Ca^{++} exchanger (NCX1) extrudes Ca^{++} (a bi-directional exchanger)
- ↓$[Ca^{++}]_{Pl}$ → low NCX1 activity → Ca^{++}-ATPase is the major pathway.
- NCX1 begins to make a significant contribution when ↑$[Ca^{++}]_{Pl}$
- Normal $[Ca^{++}]_{Pl}$ & EC gradient falls (e.g., ↑$[Na^+]_{IC}$) → NCX1 reverse direction → ↑$[Ca^{++}]_{IC}$







Renal Regulation of Mg^{++}

- Body maintains the total $[Mg^{++}]_{Pl}$ w/i narrow limits (0.8-1.0 mM)
- ~99% of the total body stores of Mg^{++} reside within:
  - Bone (~54%)
  - I.C compartment (~45%)
- Not chemically linked with other components of hydroxyapatite.
- Filterable Portion = Ionized Mg^{++}+ Diffusible Mg^{++} complexes (PO_3^{-2}, citrate, oxalate)
- Disturbances of Mg^{++} metabolism usually involve abnormal losses
- Occur most frequently during GI malabsorption, diarrhea, renal disease & diuretic use.
- Clinical manifestations:
  - Neurological disturbances (e.g., tetany (especially when associated with hypocalcemia), cardiac arrhythmias, ↑ PVR.
- Increased Mg^{++} intake:
  - ↓BP
- Severe hypermagnesemia, following excessive intake or renal failure, results in a toxic syndrome involving nausea, hyporeflexia, respiratory insufficiency, and cardiac arrest.

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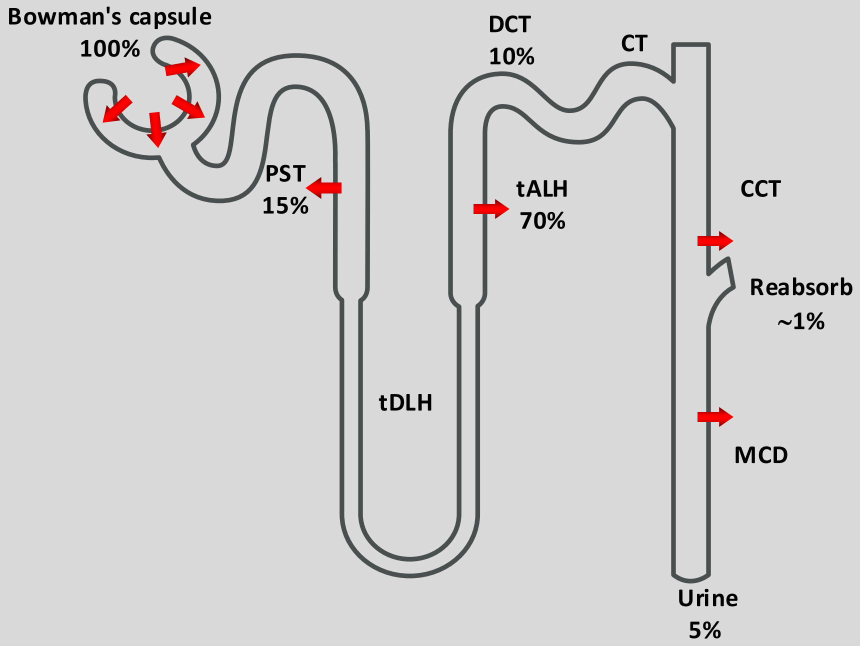
Mg^{++} Homeostasis in the Nephron



	mg/dL	mM	% Total
Ionized Mg^{++}	1.3	0.56	62
Diffusible Mg^{++} complexes	0.1	0.06	7
Protein-bound magnesium	0.6	0.28	31
Total magnesium	2.0	0.90	100

Mg^{++} Transport in the Nephron

- In contrast to the predominantly PT reabsorption pattern of ultrafiltrate, majority of Mg^{++} reabsorption occurs along tALH.

PT:

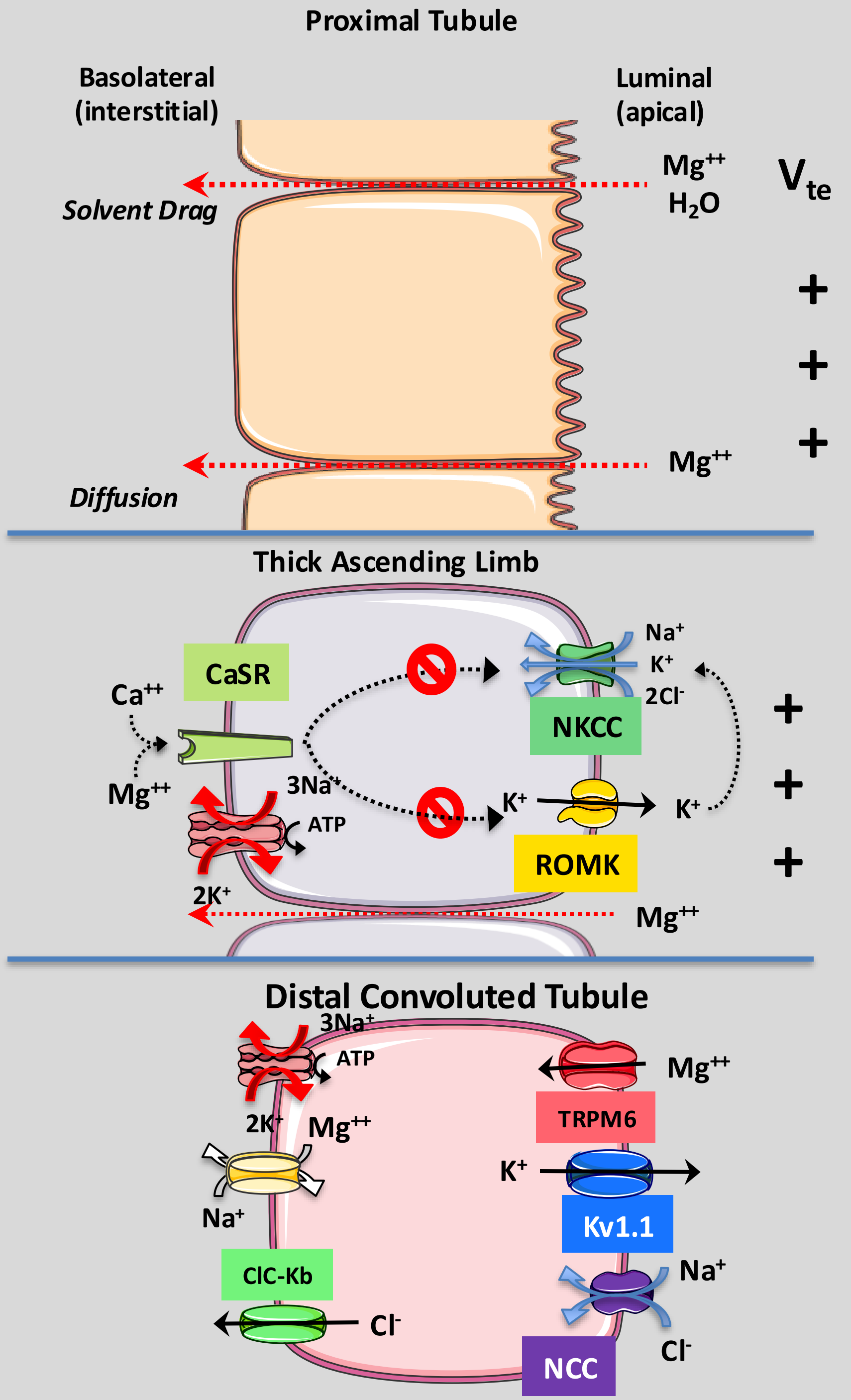
- Reabsorbs ~15% of the FL of Mg^{++}.
- H_2O reabsorption in the PT causes $[Mg^{++}]_L$ to 2x vs $[Mg^{++}]_{BC}$ Bowman's space
- Favors $[Mg^{++}]$ EC gradient for paracellular reabsorption.
- Solvent drag contributes to paracellular reabsorption.

tALH:

- Reabsorbs ~70% of the FL of Mg^{++}.
- Lumen-positive V_{te} is driving force for paracellular reabsorption.
- Tight-junction proteins claudin 16 & 19 (CLDN16 & CLDN19 genes) account for the high paracellular permeability.
- CLDN16 or CLDN19 gene mutations cause an autosomal recessive disorder of renal Mg^{++} wasting.

DCT:

- Reabsorption along the DCT is predominantly transcellular
- ~10% of the FL of Mg^{++}.
- K^+ efflux neutralizes the entry of positive charge via Mg^{++} uptake.
- Reabsorption is also stimulated by activation of epidermal growth factor receptor (EGFR).



Regulation of Mg^{++} Reabsorption

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- Mg^{++} reabsorption is stimulated by the epidermal growth factor receptor (EGFR).
- Renal Mg^{++}-wasting syndromes:
  - Mutations in TRPM6, Kv1.1, and EGF.
- Effective Circulating Volume (ECV):
  - ↓ECV → ↑Na^{+} reabsorption & ↑EC force → ↑Mg^{++} reabsorption.
  - ↑ECV → ↓Na^{+} reabsorption & ↓ EC → ↓Mg^{++} reabsorption.
- Acid-Base Balance: Identical to Ca^{++}
  - Alkalosis: ↑Mg^{++} reabsorption (↓Mg^{++} excretion)
  - Acidosis: ↓Mg^{++} reabsorption (↑Mg^{++} excretion)
- Hypomagnesemia:
  - Reduction of fractional excretion of Mg^{++}
- Hypermagnesemia/Hypercalcemia:
  - Kidneys discriminate poorly between ↑$[Mg^{++}]_{Pl}$/↑$[Ca^{++}]_{Pl}$.
  - Each will ↓reabsorption of both Mg^{++} & Ca^{++} due to binding to CaSR.
- Hormones:
  - PTH: Most important hormone for Mg^{++} regulation
  - AVP, glucagon & calcitonin: ↑reabsorption (TAL)
- Diuretics:
  - In general, diuretics ↓Mg^{++} reabsorption & ↑ excretion.

Factors Affecting Mg^{++} Reabsorption

Nephron Segment	Increase	Decrease
PT	Volume Contraction	Volume Expansion
tALH	PTH, Calcitonin, Glucagon, AVP/ADH, Hypomagnesemia, Metabolic alkalosis	Loop Diuretic, Osmotic Diuretics, Hypermagnesemia, Hypercalcemia
DCT	PTH, Calcitonin, Glucagon, Estrogens, Hypomagnesemia, K^{-}-sparing diuretics	Hypermagnesemia, Hypercalcemia, Metabolic Acidosis, Hypokalemia, Hypophosphatemia Thiazide Diuretics

Renal Regulation of PO_4^{3-}

- PO_4^{3-} is no less important than Ca^{++} & plays a critical role in many cellular processes.
- Part of ATP $\rightarrow$ Critical role in cellular energy metabolism.
- $[PO_4^{3-}]_{Pl}$ is not strictly regulated.
- Levels fluctuate throughout the day, particularly after meals.
- Ca^{++} & PO_4^{3-} homeostasis are:
  - Principal components of hydroxyapatite $[Ca_{10}(PO_4)_6(OH)_2]$ (major portion of the mineral phase of bone)
  - Regulated by parathyroid hormone (PTH), 1,25-dihydroxyvitamin D (calcitriol) & calcitonin.
- Total $[HPO_4^{-2}]_{Pl}$ & $[H_2PO_4^{-}]_{Pl}$ in adults: 0.8-1.5 mM.
- ~ 50% higher in children.
- Ionized phosphates & Diffusible phosphate complexes (complexed to Na^+, Ca^{+2}, or Mg^{+2} are filtered by the kidneys
- Soft tissues phosphates: Mainly organic phosphates (e.g., PL, phosphoproteins, nucleic acids & nucleotides
- The kidneys filter ~14 times the total E.C pool of phosphate per day (~7000 mg/day) and reabsorb (~6100 mg/day)
- Net renal excretion of phosphorus is the same as the net absorption by the GI tract.

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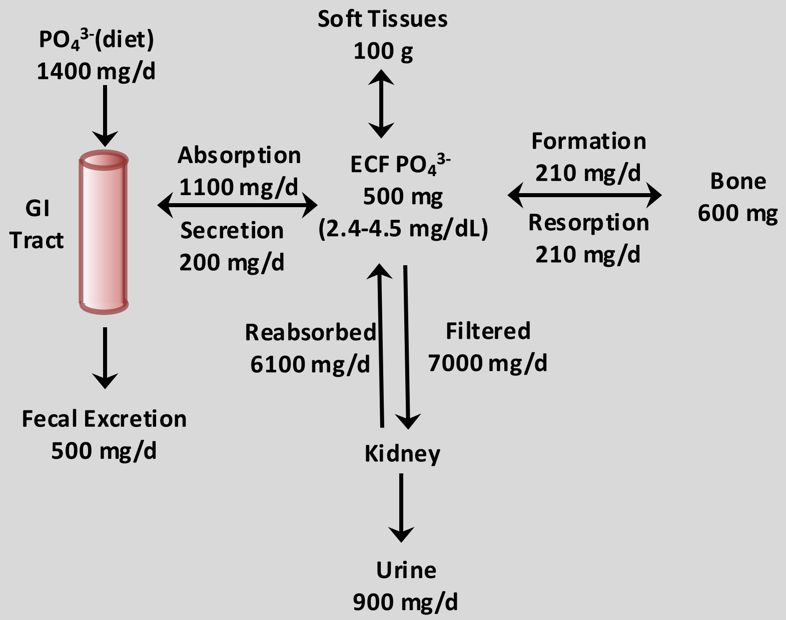
PO_4^{3-} Distribution & Balance

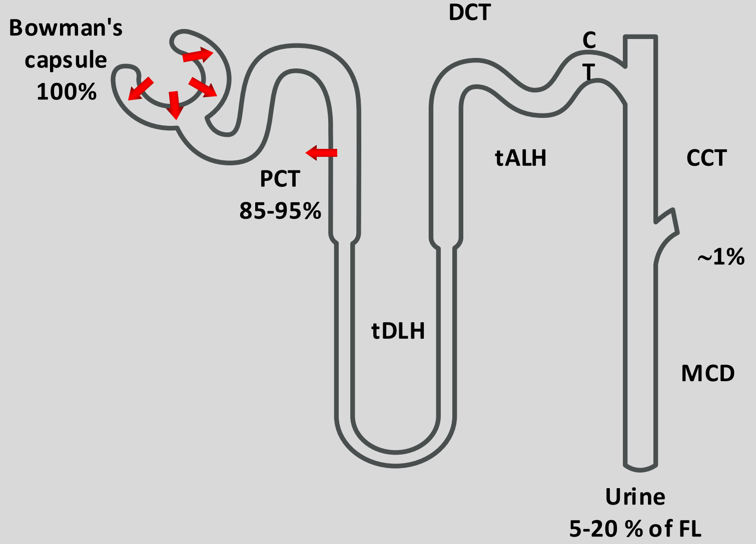


	mg/dL	mM	% Total
Ionized HPO_4^{-2} & $H_2PO_4^{-}$	2.1	0.7	50
Diffusible phosphate complexes	1.5	0.5	40
Protein-bound phosphates	0.6	0.2	10
Total phosphate	4.2	1.4	100

HPO_4^{-2} & $H_2PO_4^-$ Transport in the Nephron

- The proximal tubule reabsorbs 80% to 95% of $HPO_4^{-2}/H_2PO_4^-$ FL via transcellular route
- Distal nephron segments reabsorb a negligible fraction of the Fl.
- Three separate Na^+/phosphate cotransporters contribute: NaPi-IIa, NaPi-IIc
- Renal phosphate transport is regulated by:
  - Hormones
    - Parathyroid hormone (PTH)
    - Dopamine
    - 1,25-dihydroxy-vitamin D
    - Phosphatonins (e.g., FGF23): Circulating factors that cause renal phosphate wasting in disease (e.g., oncogenic osteomalacia, autosomal dominant & X-linked hypophosphatemic rickets).
    - Glucocorticoids
  - Acid-base balance

HPO_4^{-2} & $H_2PO_4^-$ Homeostasis in the Nephron



Proximal Tubules